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RESEARCH ARTICLE

LVADet: A Computer Vision-Based System for Detecting Lymphatic Vessels in Indocyanine Green Images

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ABSTRACT Lymphaticovenous anastomosis (LVA) is a critical surgical approach for managing limb lymphedema, proving particularly effective in mild cases where lymphatic vessels are visible via indocyanine green (ICG) imaging. However, in advanced cases, substantial lymphatic obstructions manifest as stardust patterns in ICG images, posing diagnostic challenges. Traditionally, surgeons have relied on manually extrapolating the position and orientation of lymphatic vessels from the unaffected limb to the affected one based on the assumption of bodily symmetry. To streamline this process and enhance accuracy, we developed the LVA detection model (LVADet), which employs computer vision-based artificial intelligence techniques to automate the detection of lymphatic vessels. LVADet uses a segmentation algorithm to identify lymphatic vessels on ICG images of the unaffected limb and performs keypoint detection on both limbs. Through Procrustes analysis, keypoints from both limbs are used to construct an affine transformation matrix, enabling the projection of segmented vessels from the unaffected limb onto the image of the affected limb. We tested the model using a dataset of 52 lymphedema cases, with 42 used for training and 10 for testing. This included high-quality annotations of segmentations, keypoints, and incisional wounds. Our findings, supported by rigorous evaluation and ablation studies, validate the model's design choices. Furthermore, we integrated LVADet into a prototype system using Microsoft HoloLens, facilitating real-time identification of lymphatic vessels during LVA procedures. This integration significantly reduces labor and time in the preparatory phases of surgery and shows the potential to mark a substantial advancement in surgical treatments for lymphedema.

INDEX TERMS Computer vision, lymphatic vessels, lymphaticovenous anastomosis, machine learning.

I. INTRODUCTION

Lymphedema arises when parts of the lymphatic system sustain damage or become obstructed, leading to the accumulation of lymph fluid within tissues and subsequent swelling. Various factors, such as infections, surgical procedures, and radiation therapy, can trigger this condition. Clinically, limb lymphedema manifests as edema, sensations of heaviness, and intermittent pain in the affected area. In advanced stages, limb lymphedema may lead to skin changes including discoloration, verrucous hyperplasia, hyperkeratosis, and papilloma, potentially culminating in elephantiasis. A variety of therapeutic strategies are available for treating lymphedema, which range from exercise and dietary modifications to the use of custom compression sleeves or elastic bandages within a physical therapy framework.

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Surgical options for managing lymphedema include vascularized lymph node transfer [1], [2], [3], [4] and lymphaticovenous anastomosis (LVA) [5], [6].

Lymphaticovenous anastomosis (LVA) represents a surgical intervention for lymphedema, wherein lymphatic vessels are anastomosed to an adjacent vein to bypass obstructed lymphatic regions. The objective is to reroute lymph fluid accumulation from the tissue back into the vascular system. Existing research has demonstrated the efficacy of LVA in managing early-stage limb lymphedema [7], [8], [9], [10], [11]. The success of LVA largely depends on the quality of indocyanine green (ICG) imaging. Before surgery, approximately 0.5 mL of an ICG solution (composed of 25 mg of ICG dissolved in 10 mL of water) is subcutaneously administered into the webs of the toes or fingers of the extremities. Ordinarily, the ICG migrates retrograde from the lymphatic vessels. The Fluobeam clinical system (FLUOBEAM®, France), a near-infrared fluorescence imaging system, is subsequently used to visualize the ICG. This imaging process facilitates the identification of the positions and orientations of lymphatic vessels, as shown in Fig. 1(a).

However, in cases of advanced limb lymphedema, the injected ICG may disperse across the entire limb, complicating the accurate determination of lymphatic vessel positions and orientations in the resultant ICG images, as depicted in Fig. 1(b). This diffusion can adversely affect the success of the LVA procedure. To address these challenges, current clinical protocols for LVA [12], [13] rely on approximating the distance between critical points, such as from an ankle joint to a lymphatic vessel, based on the assumption of human body symmetry. Additionally, surgeons assess the circumference of each limb to project the locations of lymphatic vessels in a limb affected by lymphedema. The preparatory procedures for LVA are illustrated in Fig. 1(c). This process demands significant effort from the medical team.

To minimize manual effort, we developed a computer vision-assisted detection model named LVADet that automates the detection of lymphatic vessels in ICG images of lymphedematous limbs. LVADet then maps the positions of lymphatic vessels from an ICG image of an unaffected limb to an ICG image of an edematous limb, as illustrated in Fig. 1(d), utilizing the assumption of human body symmetry, a conventional approach in LVA [12], [13]. To implement this concept, we introduce a three-stage framework. In the first stage, the Attention U-Net [14] deep neural network segments lymphatic vessels in an ICG image from the unaffected limb of a patient with lymphedema. In the second stage, the keypoint region-based convolutional neural network (R-CNN) model, derived from the Faster and Mask R-CNN architectures [5], identifies the position of the unaffected foot in the ICG image through keypoint and bounding box detection. In the third stage, Procrustes analysis [15] is employed to compute an affine transformation matrix, which enables the projection of identified keypoints from the ICG image of the unaffected foot onto the ICG image of the affected foot. This framework demonstrates promising results in accurately projecting computed lymphatic vessels onto the target limbs.

We assessed our proposed method using a dataset comprising 52 cases of lymphedema, each annotated with multiple segments of lymphatic vessels, foot bounding boxes, keypoints, and incisional wounds. These annotations were utilized to train LVADet. We established various evaluation protocols to assess the effectiveness of the proposed framework. LVADet proved capable of accurately identifying lymphatic vessels. Various ablation studies were performed to verify the design choices of the proposed framework. Additionally, annotations of incisional wounds were used to evaluate the inferred positions of lymphatic vessels. Given that surgical experience suggests the optimal incision site is typically located on the dorsal foot at the juncture where lymphatic vessels converge with the ankle linkage, we visualized the outcomes of LVADet and proposed incision positions, comparing these with the labeled wounds. This qualitative analysis underscores its potential utility in assisting surgeons.

We have implemented LVADet on the Microsoft HoloLens to assess its functionality. Utilizing the HoloLens, surgeons are able to view an overlay of the lymphatic vessel mapping mask on the RGB image of the patient's foot, thereby enhancing the precision and efficiency of surgical planning. This advanced visualization technique not only improves the accuracy of surgical procedures but also diminishes the time required for preparation, thereby boosting overall surgical efficiency.

In summary, this study introduces LVADet, a computer vision-assisted detection model that automates the detection of lymphatic vessels in ICG images of lymphedematous limbs. Utilizing a three-stage framework that includes segmentation, keypoint detection, and Procrustes analysis, LVADet accurately maps lymphatic vessel positions from unaffected to affected limbs. Furthermore, its integration with an augmented reality device underscores its capability to enhance surgical planning substantially, representing a significant advancement in the treatment of limb lymphedema.

II. RELATED WORKS

A. LITERATURE ON LVA AND DIAGNOSIS WITH ICG IMAGES

Limb lymphedema [16] frequently emerges as a complication following cancer treatments, which may involve lymph node dissection, chemotherapy, and radiotherapy. LVA [5] serves as a highly effective supermicrosurgical intervention for this condition. It is essential to minimize incision length and optimize efficiency in the application of LVA for lymphedema treatment, necessitating the prior identification of the orientations and locations of lymphatic vessels. This is typically achieved through indocyanine green (ICG) lymphography, which allows surgeons to visualize the lymphatic network before surgery [17], [18], [19]. Recent advancements have explored the use of computer vision and artificial intelligence

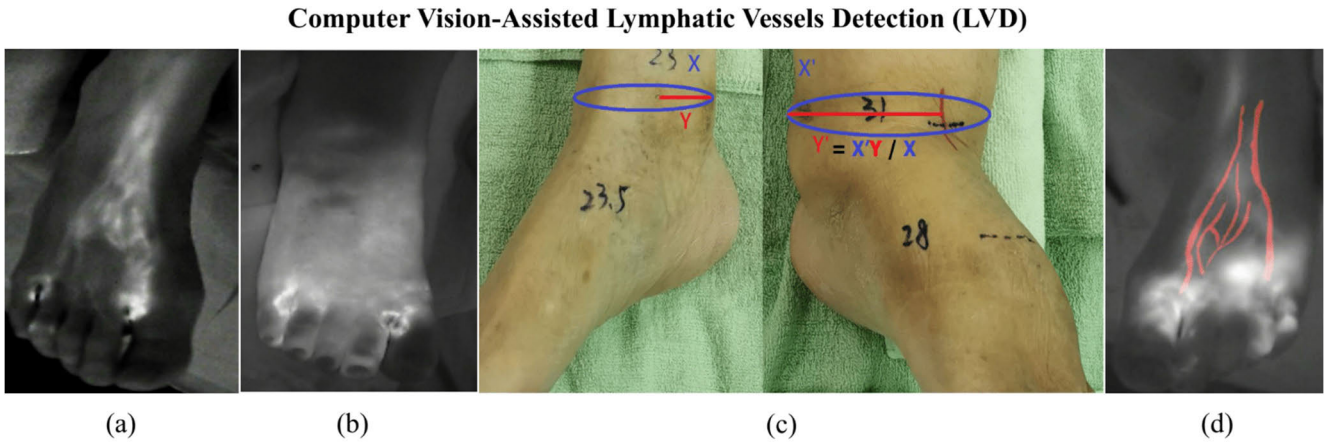

FIGURE 1. (a) An ICG image of a unaffected foot. (b) An ICG image of a foot with lymphedema. (c) Current protocols for LVA necessitate the manual determination of lymphatic vessel positions by the surgical team. This involves measuring the circumference of a normal limb (X) and a lymphedematous limb (X′), as well as measuring the distance from a reference point (e.g., the medial malleolus) to a lymphatic vessel [Y; as identified on an indocyanine green (ICG) image], and the distance from the same reference point to the predicted position of the lymphatic vessel (Y′). The formula used is X: X′ = Y: Y′ (d) Process of lymphatic vessel detection, which involves mapping the coordinates of lymphatic vessels from the ICG image for an unaffected foot to an ICG image for a foot with lymphedema.

techniques to automate the interpretation of ICG image. For instance, Ozmen et al. [20] proposed a deep learning-based classification system for analyzing ICG lymphography patterns to aid in the diagnosis of lymphatic diseases.

However, precisely identifying the locations of lymphatic vessels in advanced limb lymphedema is challenging due to the presence of stardust patterns in the affected limb on ICG images [Fig. 1(b)]. The orientations and locations of lymphatic vessels in limbs with advanced lymphedema can be ascertained either manually through surgical experience or automatically via symmetric mapping techniques. Traditional methods rooted in surgical experience or empirical approaches, such as the superior-edge-of-the-knee incision technique [12], [21], [22], entail locating viable lymphatic vessels near the great saphenous vein for LVA. Conversely, symmetric mapping techniques, including contralateral AR mapping [12] and predictive lymphatic mapping [13], offer greater precision than manual methods. While this method can offer more precision, it is time-consuming and labor-intensive.

To overcome these limitations, our approach leverages advanced computer vision and artificial intelligence techniques to automate the processing of ICG images, facilitating a more efficient and accurate mapping of lymphatic vessels. By integrating segmentation, keypoint detection, and Procrustes-based mapping, our method reduces the need for manual calculations and accelerates the preparatory phases of LVA, offering a practical solution for improving surgical outcomes.

B. SEGMENTATION OF MEDICAL AND BIOMEDICAL IMAGE

Encoder-decoder models, such as U-Net [23] and V-Net [24], are commonly adopted for the segmentation of medical and biomedical images. U-Net won the ISBI Cell Tracking Challenge 2015, and V-Net was used for the three-dimensional (3D) segmentation of magnetic resonance imaging (MRI) volumes depicting the prostate. Based on this encoder-decoder architecture, several works have been developed to enhance segmentation performance in various medical imaging applications. For instance, Attention U-Net [14], which is composed of attention gates and U-Net, is commonly used for the multiclass segmentation of abdominal computed tomography (CT) images. Progressive Dense V-Network [25] enables the reliable and automatic segmentation of lung lobes, thus aiding in pulmonary disease diagnosis.

In addition to these applications, several studies have focused on improving the segmentation of vascular structures using U-Net-based architectures. These approaches have been successfully applied to various types of vascular segmentation, including brain vessel segmentation [26], [27], [28], retinal vessel segmentation [29], [30], [31], hand and finger vein segmentation [32], [33], liver vascular segmentation [34], and lymphatic vessel segmentation [35]. Such studies underscore the versatility and effectiveness of U-Net-based methods in automating the segmentation of complex vascular structures.

C. KEYPOINT DETECTION AND PROCRUSTES ANALYSIS

Keypoints play a crucial role across various applications. Human-pose-related keypoints [36], [37], [38] identify the joints and limbs of an individual, facilitating precise human pose estimation. Hand-pose-related keypoints [39] are employed to interpret human actions and to compile sign language datasets. Clothing-related keypoints [40], [41], [42] aid in detecting clothing boundaries and in analyzing fashion trends. Facial keypoints [43] serve as the foundation for facial recognition and emotion analysis. Foot-related keypoints are

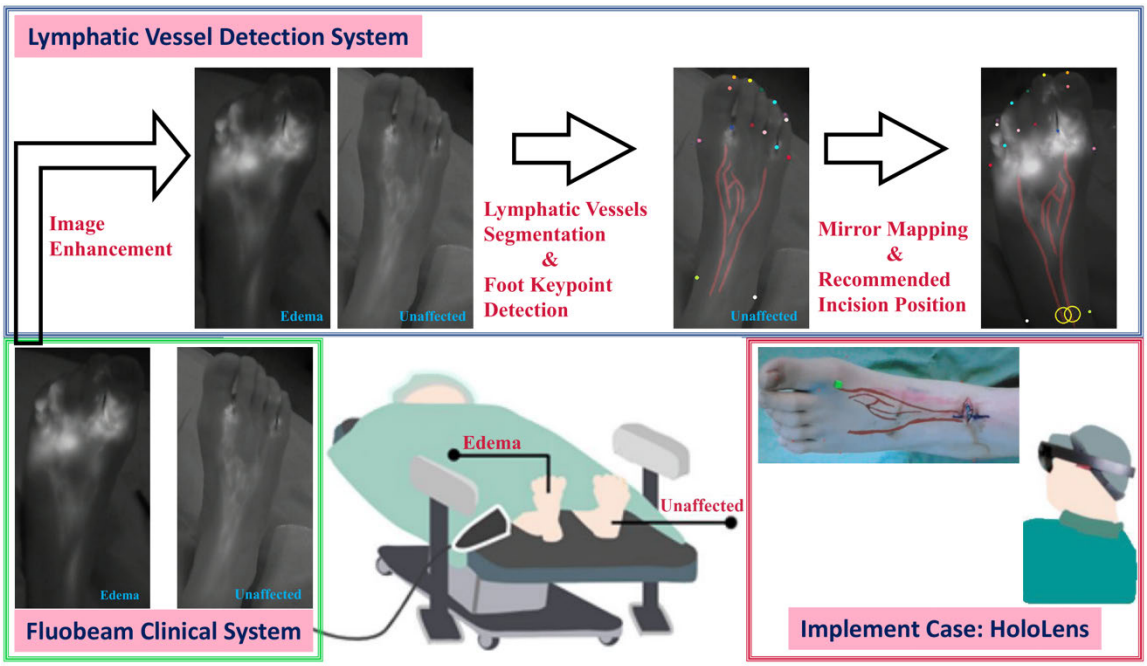

FIGURE 2. Processing of ICG images by the developed computer-vision-assisted LVA detection model (LVADet). The Fluobeam clinical system (FLUOBEAM®, France) is used to obtain ICG images of healthy and lymphedema-affected feet through near-infrared fluorescence imaging. LVADet conducts image enhancement, lymphatic vessel segmentation, foot keypoint detection, and mirror mapping. A lymphatic vessel mask is obtained from the ICG image of the normal foot of a patient with lymphedema, and foot keypoints are detected using the keypoint region-based convolutional neural network (R- CNN). The obtained keypoints and mask are then used to implement mirror mapping through Procrustes analysis. The results of LVADet can be combined with surgeon experience to identify a suitable incision position. The implementation of LVADet on Microsoft HoloLens is also illustrated in the figure.

used to outline the foot contour, which is crucial not only for customized shoe design [44] but also for various medical applications. For example, detecting foot keypoints has been applied in foot monitoring and serves as a valuable assessment tool for the early identification of dementia disorders [45].

Procrustes analysis [15] involves examining the distribution of a set of shapes to determine their relationships and consistency. This is done using the least squares approach to determine the affine transformation from shape A to shape B, including the translation, scaling, and rotation components of this transformation. Procrustes analysis not only plays a crucial role in face recognition and matching [46] but also is suitable for analyzing biological data, such as data related to tooth shape [47].

D. AR METHODS FOR SURGERIES

Wearable AR devices, such as HoloLens, can support surgical operations and rehabilitation treatment by offering different functionalities, such as visualization, immersion, and interaction. Visualized medical data are used in AR systems [48], vascular localization systems [49] and glenoid drilling guides [50] to assist surgeons. The immersiveness of AR is useful for pathology [51], endoscopic surgery [52], and the visualization of MRI data [53]. The interactivity of AR is useful for holographic pedicle screw navigation [54], CT-guided lesion targeting [55], automatic flexible endoscope control [56], and balance rehabilitation [57].

III. METHOD

We propose LVADet, a computer-vision-assisted lymphaticovenous anastomosis detection model, aimed at projecting lymphatic vessels from unaffected to lymphedematous limb. This methodology encompasses three key stages: lymphatic vessel segmentation, foot keypoint detection, and mirror mapping. First, a deep neural network segments lymphatic vessels in an ICG image of the unaffected foot. Next, a separate neural network detects foot keypoints in both unaffected and lymphedematous foot images. Finally, using these keypoints, mirror mapping algorithms transpose the lymphatic vessel structure from the unaffected foot to the lymphedematous foot. Fig. 2 illustrates the workflow of this proposed method.

A. DATA PREPARATION

To construct a high-quality LVA dataset, we defined three criteria for assembling images. Firstly, selection of ICG images depicting the entire foot is required. Secondly, lymphatic vessels on the unaffected limbs should be visible and distinct in the ICG images. Thirdly, all defined keypoints on the foot should be identifiable in the images. Following the selection process, we implemented affine transformations to ensure

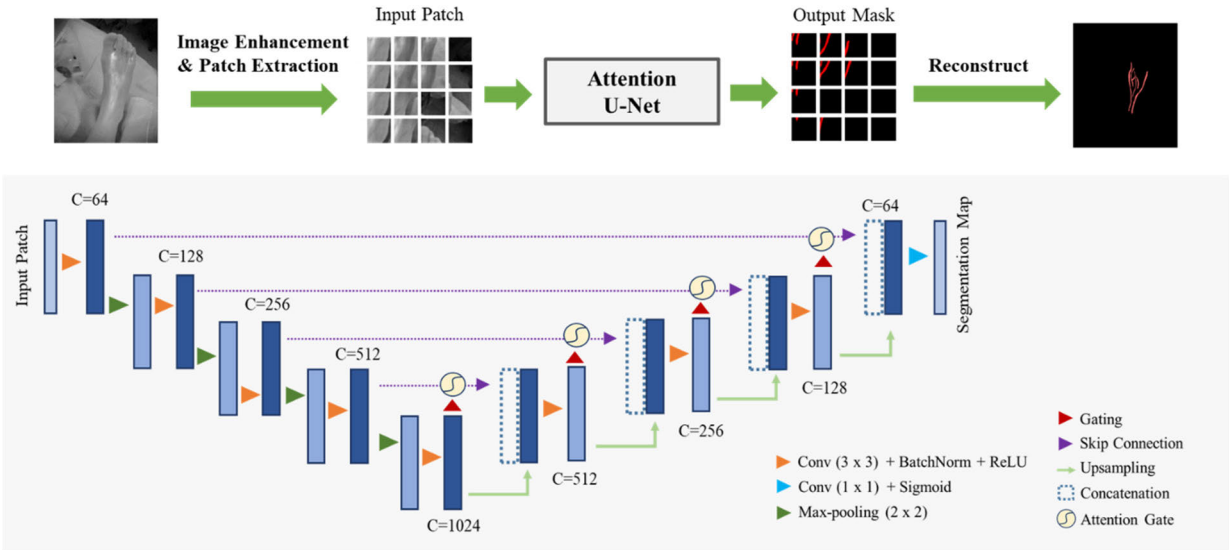

FIGURE 3. The top section shows the process of the patch-based segmentation algorithm, while the bottom section displays the network architecture of the Attention U-Net model [14].

consistent alignment of all limbs in the ICG images, maintaining vertical positioning. Each case is categorized as either mild or advanced, reflecting varying levels of distortion and the corresponding difficulty in identifying lymphatic vessels.

B. LYMPHATIC VESSEL SEGMENTATION

We utilized supervised learning to train a deep neural network for the segmentation of lymphatic vessels. To improve detection accuracy with a limited dataset, we implemented patch-based learning, which entails extracting patches from the full image and utilizing these patches as training inputs for the segmentation model. These patches are then reconstructed to their original sizes for inference and visualization. The architecture of the employed segmentation model is elaborated in Fig. 3.

1) PATCH EXTRACTION

In our approach, we employed patch-based learning for both training and testing of the model. To compile a training dataset, we extracted a sufficient number of patches from labeled data, which were then utilized to update the model parameters. For testing the model, we applied an overlapping extraction strategy to derive patches from the test data. These overlapping patches served as inputs for the segmentation model, and the outputs were subsequently reconstructed to generate the final results. The methodologies used for extraction and reconstruction are elaborated by C. Wang et al [31]. These techniques were implemented in the context of patch-based learning for retinal vessel segmentation.

In our methodology, the bounding box delineated by a foot detection model is employed to exclude background information prior to patch extraction. The input images used in this study were 576 × 696 pixels in size, from which we extracted patches measuring 64 × 64 pixels during the training phase. This procedure enabled the expansion of the dataset to include approximately 100 000 patches. Additionally, we conducted normalization and applied standard image augmentation techniques, including flipping, rotation, and brightness adjustment, to improve contrast and diversify the data.

2) ATTENTION U-Net ARCHITECTURE

U-Net is commonly used in medical image segmentation. The notable characteristic of this model is its capacity to merge high- and low-level features, which results in it having excellent performance even for limited datasets. In this study, we employed a variant of U-Net called Attention U-Net [14]. Attention U-Net was created by incorporating the attention mechanism expressed in (1) into the basic U-Net framework to suppress irrelevant regions within images while highlighting important features in specific areas. This model exhibits higher sensitivity and prediction accuracy than does the basic U-Net model. In (1), ψ, W_x and W_g represent convolution operations; x^l represents the input features; g is the gating signal; b_ψ and b_g are convolution biases; σ_1 is the rectified linear unit activation function; σ_2 is the sigmoid activation; and $\alpha_i \in (0, 1]$ is an attention coefficient.

$$q_{att} = \psi^T \left(\sigma_1 \left(W_x^T x_i^l + W_g^T g_i + b_g\right)\right) + b_\psi$$
$$\alpha_i^l = \sigma_2 \left(q_{att}^l \left(x_i^l, g_i; \theta_{att}\right)\right) \quad (1)$$

We used image patches as input for the Attention U-Net model and employed cross-entropy as the loss function to update the model parameters. The pixel-wise cross-entropy loss is expressed in (2). The t_i is the ground truth label for pixel i, and the p_i is the predicted probability for pixel i. Training was conducted with an initial learning rate of 0.0001 for

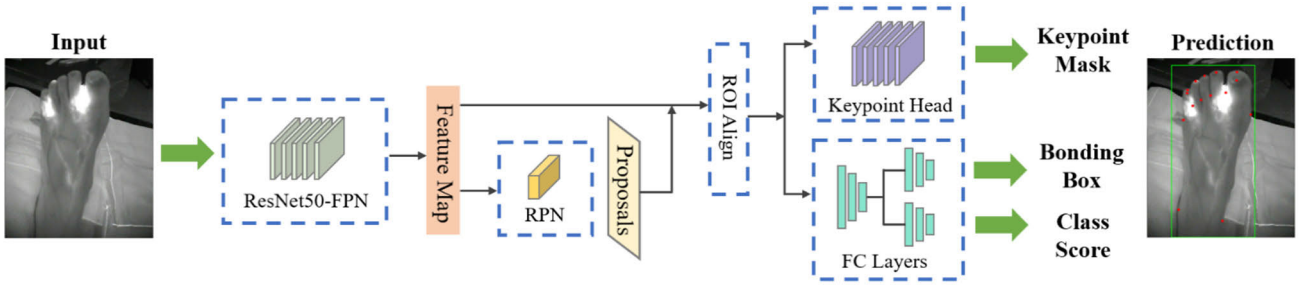

FIGURE 4. The architecture of the keypoint R-CNN.

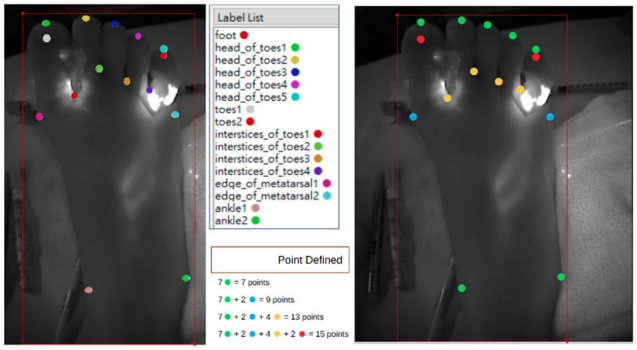

FIGURE 5. The delineation of keypoint labels (Left) and the varying numbers of keypoints utilized in our experiments, as presented in Table 3 (right).

30 epochs, and the Adam optimizer was used for optimization.

$$L_{seg} = -\sum_{i=1}^{n} t_i \log(p_i) \tag{2}$$

C. FOOT KEYPOINT DETECTION

To enable the projection of output lymphatic vessel masks onto the target foot, an additional deep neural network was trained to autonomously predict the locations of crucial foot keypoints. We designated 15 foot keypoints: one at the tip of each toe (five points in total), one at the base of the big and little toes (two points in total), four at the midpoints between adjacent toes, and four at the junctures of the metatarsals and the ankle (Fig. 4). In our LVA detection system, we utilized the keypoint R-CNN model for detecting foot keypoints. The keypoint R-CNN is a deep learning model that extends the capabilities of the Faster [58] and Mask R-CNN [5] models, originally developed for automatic detection of human-pose-related keypoints. Key processes in this model include region proposal network (RPN) [59], [60] and the Region of Interest (ROI) Align algorithm. Initially, input images undergo processing through region proposal layers that generate potential object regions. These regions are then refined and classified in subsequent stages of the network. These regions are subsequently refined and categorized in later stages of the network. The extracted feature maps are finally analyzed by the keypoint head, which determines the positions of specific keypoints. Fig. 5 depicts the comprehensive architecture of the keypoint R-CNN model, with ResNet-50-FPN [61], [62] serving as the backbone for feature extraction.

During the training of the keypoint RCNN model, the multi-purpose loss functions encompass classification, keypoint, bounding box regression, RPN classification, and RPN localization losses, as detailed in (3):

$$L_{total} = L_{cls} + L_{kpt} + L_{box_{reg}} + L_{rpn_cls} + L_{rpn_loc} \tag{3}$$

The object classification loss (L_{cls}) and RPN classification loss (L_{rpn_cls}) are both calculated using cross-entropy. Here t_i represents the ground truth label for class i, and p_i denotes the predicted probability for class i.

$$L_{cls} = -\sum_{i=1}^{n} t_i log(p_i) \tag{4}$$

The keypoint loss (L_{kpt}) is calculated using L1 loss. In this context, v represents the predicted keypoint positions, v^* denotes the ground truth keypoint positions, and the N_k is the number of keypoints.

$$L_{kpt} = \frac{1}{N_k} \sum_{i=1}^{N_k} \left\| v_i - v_i^* \right\|_1 \tag{5}$$

The bounding box regression (L_{box_reg}) and RPN localization (L_{rpn_loc}) are computed using smooth L1 losses, where t_i and t_i^* represent the predicted and ground truth parameters of the bounding boxes, respectively.

$$L_{smooth_l1} = \sum_{i \in \{x,y,w,h\}} smooth_{L_1}\left(t_i - t_i^*\right)$$
$$smooth_{L_1}(x) = \begin{cases} 0.5x^2 & \text{if } |x| < 1 \\ |x| - 0.5 & \text{otherwise} \end{cases} \tag{6}$$

For data preprocessing and augmentation, we implemented normalization and random flipping to enhance dataset quality. The initial training parameters set for the keypoint R-CNN model included a learning rate of 0.0001, a total of 300 epochs, and the Adam optimizer. Additionally, we utilized pretrained weights from ResNet-50 on ImageNet [63] for the extraction of image features.

D. MIRROR MAPPING

Using the segmentation model, we obtained the lymphatic vessel mask from the unaffected foot. The keypoint detection model provided the coordinates of keypoints in the

ICG images of both the unaffected and affected limbs. With these keypoints, we computed an affine transformation matrix using Procrustes analysis. This matrix enabled us to project the lymphatic vessel mask from the unaffected limb onto the affected limb, thus completing the mirror mapping process. The procedures involved in the mirror mapping process are detailed in Algorithm 1.

Algorithm 1 Mirror Mapping by Procrustes Analysis

Require:
The mask of lymphatic vessels by lymphatic vessels segmentation from normal limb.
The keypoint set from normal limb by keypoint detection:
$K_n = (X_{fn1}, Y_{fn1}), (X_{fn2}, Y_{fn2}), \ldots, (X_{fn15}, Y_{fn15})$
The keypoint set from edema limb: by keypoint detection:
$K_e = (X_{e1}, Y_{e1}), (X_{e2}, Y_{e2}), \ldots, (X_{e15}, Y_{e15})$
1: Let edema limb to be the approximate mean set.
2: **repeat**
3: Align normal limb set to the approximate mean set:
4: (a) Calculate the mean value (centroid) of each set and align two sets centroid to the origin.
5: (b) Normalized the size of two sets, which means subtracting each point from its corresponding mean.
6: (c) Rotate each set to align with the newest approximate mean.
7: Calculate the new approximate mean from the aligned sets.
8: **until** The approximate mean from steps 3 and 7 are the same and return the transformation matrix.
9: Mirror mapping the mask of lymphatic vessels from normal limb to edema limb by the transformation matrix from step 8.

IV. EXPERIMENTAL RESULTS

We conducted comprehensive qualitative and quantitative analyses to evaluate performance of LVADet, demonstrating its capability to expedite and enhance the precise identification of lymphatic vessels. Moreover, we explored its potential integration with hardware devices for practical applications in surgical settings.

A. MATERIALS

ICG images and labeled data were sourced from the Division of Plastic and Reconstructive Surgery, Department of Surgery, at Taichung Veterans General Hospital in Taichung, Taiwan. Physicians at this facility administered approximately 0.5 mL of ICG into the first and fourth interdigital spaces of patients diagnosed with lymphedema. The injected ICG propagated through the lymphatic vessels of the foot and was subsequently imaged using the Fluobeam clinical system (FLUOBEAM®, France), a near-infrared fluorescence imaging system. The ICG images produced by the Fluobeam system are 696 × 576 pixels in size. Doctors employed the LabelMe [64] tool to annotate the data for segmentation and keypoint detection purposes. The dataset assembled includes ICG images from 52 patients. Of these patients, 30 exhibited mild lymphedema, and 22 presented with advanced lymphedema. Within this study, data from 42 patients were utilized for model training, while the data from the remaining 10 patients were allocated for model testing.

- **Lymphatic Vessel Segmentation:** The segmentation dataset used comprises images of unaffected feet with annotated lymphatic vessels, serving as templates for mapping the distribution of lymphatic vessels in limbs impacted by lymphedema. From this dataset, 248 ICG images were employed for training, while 66 ICG images were allocated for testing purposes.
- **Foot Keypoint Detection:** The foot keypoint dataset employed comprises ICG images of both unaffected feet and feet with lymphedema. Within this dataset, 719 ICG images were utilized for training purposes, consisting of 549 images of unaffected limbs and 170 images of limbs affected by lymphedema. Additionally, 134 images were allocated for testing, which included 90 images of unaffected limbs and 44 images of limbs with lymphedema.
- **Mirror Mapping:** ICG images from 30 patients with mild lymphedema were used to conduct mirror mapping through Procrustes analysis, ensuring clear and reliable results. Statistical data from the ICG images of 20 patients were utilized for training, while data from the ICG images of the remaining 10 patients were allocated for testing.

The experiments in this study were carried out on a personal computer equipped with an Ubuntu 18.04.6 LTS 64-bit operating system, an Intel Core i7-10700 CPU, and an NVIDIA GeForce RTX 3090 GPU.

TABLE 1. Evaluation metrics used in this study.

Task	Metric
Vessels Segmentation	Accuracy (ACC) = $\frac{TP+TN}{TP+FP+TN+FN}$
	Sensitivity (SE) = $\frac{TP}{TP+FN}$
	Specificity (SP) = $\frac{TN}{TN+FP}$
Keypoint Detection	Object Keypoint Similarity (OKS) $= \frac{\sum_i exp^{\frac{-d_i^2}{2s^2k_i^2}} \delta(v_i > 0)}{\sum_i \delta(v_i > 0)}$
Mirror Mapping	Procrustes Distance (D_p) = $\sqrt{\sum d_i^2}$

B. EVALUATION METRICS

The evaluation metrics for each stage are listed in Table 1.

The performance metrics used to assess the segmentation model include accuracy, area under the receiver operating characteristic curve (AUC), sensitivity, and specificity, with each metric ranging from 0 to 1. TP, TN, FP, and FN represent the counts of true positives, true negatives, false positives, and false negatives, respectively. True positives are cases where both the prediction and the ground truth are positive, true negatives are cases where both are negative, false positives occur when the prediction is positive but the ground truth is negative, and false negatives occur when the prediction is negative but the ground truth is positive. Accuracy reflects the overall correctness of the model, while AUC provides a comprehensive measure of performance across all possible classification thresholds. Sensitivity and specificity measure the proportion of actual positives and negatives, respectively, that are correctly identified by the model.

Object keypoint similarity (OKS) [65] is a widely used metric for assessing the accuracy in tasks such as human pose estimation [66] and hand keypoint detection [67]. The formula for OKS is detailed in Table 1. In this formula, d_i refers to the Euclidean distance between the ground truth and the predicted keypoint i, s indicates the actual scale of the object and the square root of the object segment area, k is a constant that controls the fall off for keypoint i, and v_i represents the ground-truth visibility flag for keypoint i. The v_i values of an unlabeled keypoint, a labeled but invisible keypoint, and a labeled and visible keypoint are 0, 1, and 2, respectively. Finally, δ refers to the Dirac-delta function, which is 1 if keypoint i is labeled and 0 otherwise. For evaluation, average precision (AP) and average recall (AR) for the Object Keypoint Similarity (OKS) are computed. AP quantifies the accuracy of keypoint detection across various Intersection over Union (IoU) thresholds, whereas AR assesses the capability to accurately identify keypoints across different levels of recall. Both AP and AR metrics are designed to achieve higher values, indicating superior performance, with the superscript values specifying the range of OKS utilized for evaluation.

In mirror mapping stage, Procrustes distance (D_p) [68] quantifies the similarity between shapes, which refers to the keypoint sets representing foot landmarks. This distance is computed as the sum of squared differences between point sets A and B, which directly affects the accuracy of projecting the lymphatic vessel mask from the unaffected foot to the affected foot. A smaller Procrustes distance denotes a greater similarity between the shapes.

C. RESULT

The results for segmentation, keypoint detection, and Procrustes analysis are outlined below. For each module, we conducted a series of experiments to assess the performance of each task using common evaluation metrics. Additionally, we include visualizations to better illustrate the outcomes of our methods, offering a clearer understanding of how each approach performs in practice.

TABLE 2. Segmentation results.

Patch	BBox	Pre	ACC	AUC	SE	SP
✗	✗	Normalize	**0.9925**	0.9879	0.6316	**0.9962**
✓	✗	Normalize	0.9845	**0.9912**	0.7371	0.9821
✓	✓	-	0.9866	0.991	0.7354	0.9919
✓	✓	Normalize	0.9851	0.9909	**0.7705**	0.9897

TABLE 3. Results for keypoint detection.

#Keypoints	AP	AP^{50}	AP^{75}	AR	AR^{50}	AR^{75}
7 points	47.4	61.4	46.3	64.8	75.4	64.2
9 points	57.7	68.4	60.3	71.6	79.9	73.1
13 points	73.1	80.8	73.7	83.3	88.8	83.6
15 points	**88.0**	**94.7**	**87.0**	**92.0**	**96.3**	**91.0**

1) LYMPHATIC VESSEL SEGMENTATION

Image segmentation was performed using the Attention U-Net model [14] which facilitated the extraction of lymphatic vessel masks in an unaffected leg. Table 2 details the segmentation outcomes under various conditions involving patch extraction, background filtering using foot bounding boxes, and image normalization. The inclusion of patch-based learning and foot bounding box filtering improved the model's ability to focus on relevant features and reduce background noise. Normalization standardized image intensities, enhancing the model's consistency and robustness. The configuration with all conditions was chosen due to its balanced performance, achieving the highest sensitivity of 0.7705, which is crucial for detecting true positives, and maintaining high specificity at 0.9897, thus minimizing false positives. Additionally, it achieved high accuracy of 0.9851 and an AUC of 0.9909, demonstrating its effectiveness.

2) FOOT KEYPOINT DETECTION

Table 3 presents the experimental results for keypoint detection using various configurations with different numbers of labeled keypoints. The first configuration designates seven points: five at the toe's forefront and two at the ankles. Adding two points at the metatarsals' periphery increases the total to nine points. Expanding further, four points are placed at intervals between the toes, resulting in thirteen points. Finally, two additional points are specified at the big and little toes, totaling fifteen points. The configuration with fifteen points yields the best results, achieving an average precision (AP) of 88.0 and an average recall (AR) of 92.0. This indicates that a greater number of keypoints enhances the model's accuracy in detecting and localizing foot keypoints.

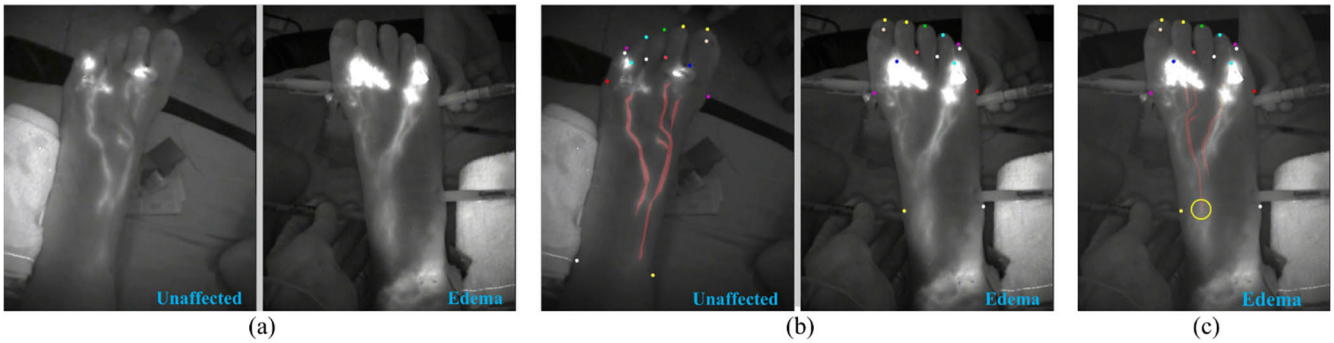

FIGURE 6. (a) Comparative analysis of ICG images displaying lymphatic vessels in an unaffected limb versus a limb with lymphedema. (b) Overlay of predicted lymphatic vessels and keypoints on ICG images for both unaffected and lymphedematous limbs using LVADet. (c) Mirror mapping outcomes derived from Procrustes analysis (the recommended incision position is indicated by a yellow circle).

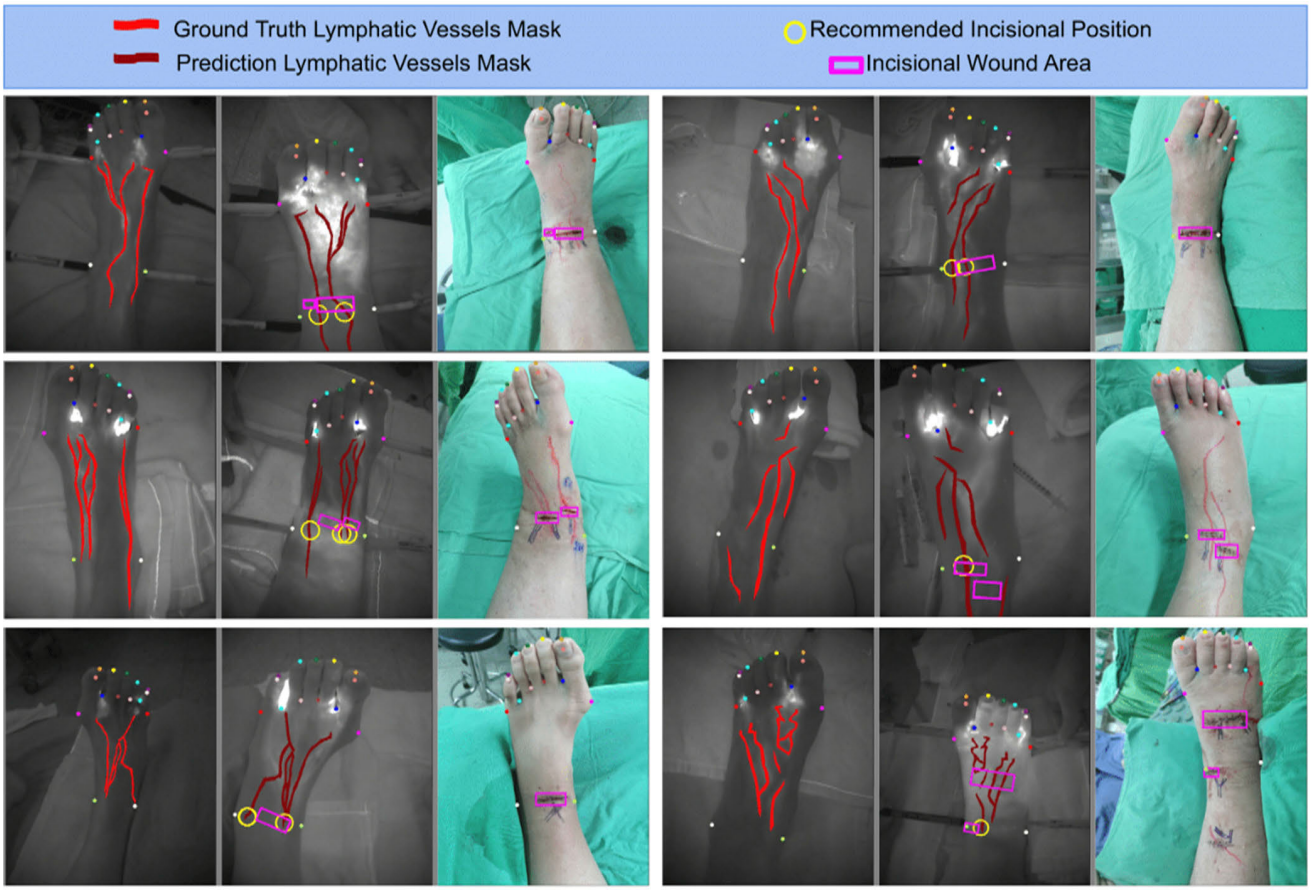

FIGURE 7. Experiment results obtained for lymphatic vessel segmentation, foot keypoint detection, mirror mapping, and incision position identification. The yellow circles indicate the recommended incision positions identified in this study on the basis of the results of LVADet and a surgeon's experience. The purple boxes indicate the locations of real wounds, which are mapped on ICG images for feet with lymphedema.

This improvement is likely due to the more detailed representation of the foot, which allows for more precise and reliable keypoint detection.

3) MIRROR MAPPING

After obtaining the lymphatic vessel mask and foot keypoints, Procrustes analysis was used to derive an affine transformation matrix for mirror mapping. This process projected the lymphatic vessel mask from an ICG image of an unaffected limb onto an ICG image of a limb with lymphedema. Fig. 6 illustrates the overall results of this process. ICG image data from 20 patients were used for training and from 10 patients for testing in the mirror mapping process. The normalized Procrustes distances for the training and

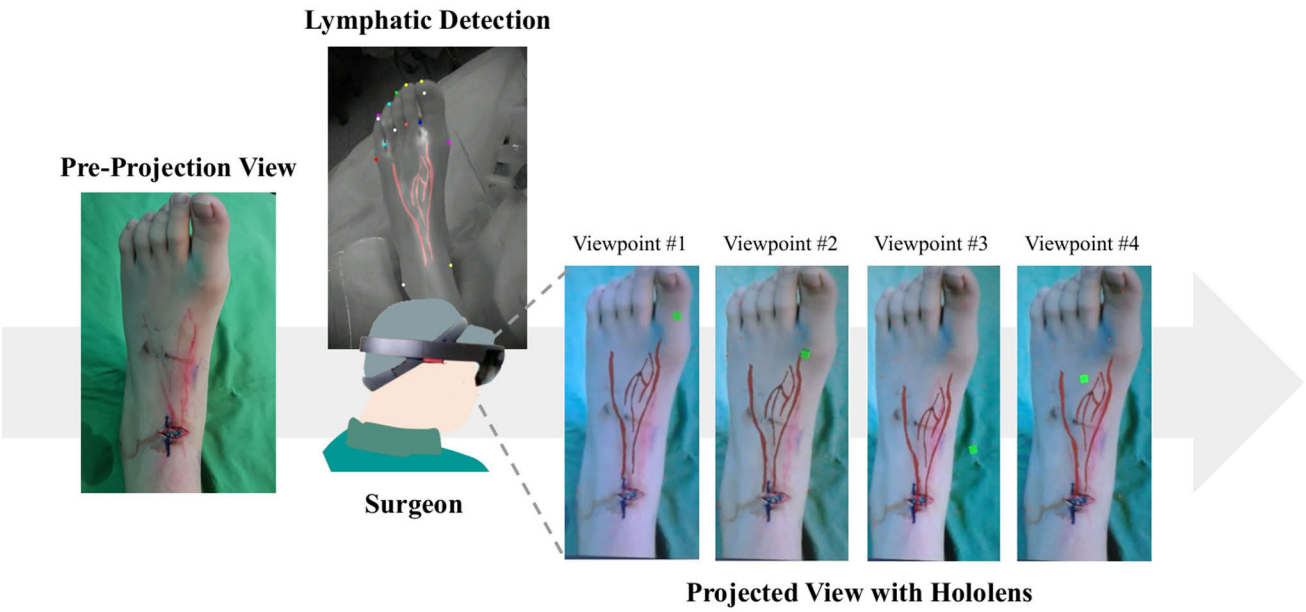

FIGURE 8. We project the LVADet results onto a patient's lymphedematous foot using the HoloLens. The left panel shows the pre-projection view, and the right panel demonstrates the projected lymphatic pathways from multiple viewpoints. This system has the potential to enhance precision and accelerate the progress of LVA surgeries.

testing datasets ranged from 0.36 to 3.85 and from 0.27 to 1.59, respectively, indicating consistent results across both datasets.

A lower Procrustes distance denotes greater similarity between the shapes delineated by the keypoints of the unaffected and lymphedema-affected feet. The distances are measured in pixels, demonstrating that the mapping results are already very close. The experimental results demonstrated that the positioning of foot keypoints significantly impacted the results of mirror mapping. The discrepancies emerged because the original foot keypoints exist in a 3D space, whereas mirror mapping is conducted in a 2D space. Additionally, although standard directions and locations for capturing ICG images were established, variations arose due to individual interpretation.

4) RECOMMENDED INCISIONAL POSITION

To evaluate the clinical applicability of LVADet, we provided mirror mapping results to a surgeon to aid in the identification of lymphatic vessels in ICG images for cases of lymphedema. These results facilitated the surgeon's ability to locate lymphatic vessels for LVA procedures. Additionally, we gathered postoperative RGB images of limbs affected by lymphedema to ascertain the wound position and assess whether it aligned with the mirror mapping results (Fig. 7). Based on surgical experience, the optimal incision site is typically located at the junction where lymphatic vessels meet the ankle linkage. Fig. 7 demonstrates that the actual wound location corresponded with the lymphatic vessel positions as determined by LVADet. Consequently, LVADet proves to be a valuable tool for assisting surgeons in pinpointing the appropriate incision location for LVA.

V. IMPLEMENTATION CASE – HOLOLENS

Lymphaticovenous anastomosis (LVA) protocols necessitate that surgeons pinpoint the positions of lymphatic vessels using the Fluobeam clinical system monitor (FLUOBEAM®, France). Both the clear visualization of lymphatic vessels in ICG images of limb lymphedema and the mapping results provided by LVADet for these images involve similar preparatory steps. To improve convenience, a lymphatic vessel detection system should allow the surgeon to simultaneously view the predicted positions of lymphatic vessels and the patient's lymphedematous foot. This requirement for concurrent visualization underscores the necessity for an augmented reality (AR) device.

We developed the LVADet model using Azure AI Custom Vision (Microsoft, Redmond, WA, USA) and deployed it on a wearable augmented reality (AR) device, specifically the first-generation Microsoft HoloLens. The implementation of LVADet involved the following steps:

Step 1: Fifty-three RGB images of feet were used to train a foot detection model on Azure AI Custom Vision, facilitating foot recognition and tracking.

Step 2: A 3D project was established in Unity, where the foot detection model was configured and linked with the Microsoft HoloLens.

Step 3: The dimensions and positioning of the predicted keypoints and lymphatic vessels were adjusted based on the bounding boxes identified in the RGB images by the foot detection model.

Step 4: The 3D Unity project was subsequently executed on the HoloLens.

Fig. 8 illustrates the results achieved using LVADet with the HoloLens, comparing the natural view with the augmented views provided at different times (the green square indicates the center of the view). With the HoloLens, physicians can view the predicted locations of lymphatic vessels and the patient's lymphedematous foot concurrently. This capability potentially accelerates the workflow of LVA surgeries.

VI. CONCLUSION

A conclusion section is not required. Although a conclusion may review the main points of the paper, do not replicate the abstract as the conclusion. A conclusion might elaborate on the importance of the work or suggest applications and extensions.

A. CONTRIBUTION

In this work, we develop a three-stage LVA detection model (LVADet), which employs image segmentation and keypoint detection to automate the identification of lymphatic vessels. This approach offers two key contributions: first, it try to address the challenge of accurately detecting lymphatic vessels in advanced lymphedema cases, where severe swelling often makes vessel identification difficult. Second, it fully automates the entire detection and mapping process, significantly improving efficiency compared to traditional methods, which involve direct patient contact, manual limb circumference measurements, and substantial manual effort. Furthermore, the integration with AR devices demonstrates a major advancement in surgical treatments for lymphedema, allowing physicians to visualize and plan incisions with greater precision and efficiency.

B. LIMITATIONS

Despite its promising outcomes, LVADet presents several limitations. Firstly, the method is confined to 2D image processing, lacking 3D spatial data. For example, ICG images captured from particular angles might introduce distortions during the transfer of detected keypoints from an unaffected limb to a limb with lymphedema in the mirror mapping phase. Additionally, certain foot regions, such as the little toe, may be obscured, resulting in biased outcomes. Secondly, LVADet's effectiveness is diminished in cases of severe lymphedema, where pronounced swelling complicates the identification of foot landmarks and the execution of mirror mapping. Moreover, LVADet is unable to identify suitable incision sites for LVA in instances of bilateral advanced lymphedema due to the absence of unaffected lymphatic vessels for reference. These scenarios remain a challenge.

C. FUTURE WORKS

We propose the following three directions to further research in this field. First, the size of the dataset used for training significantly influences the performance of supervised learning models. We strongly advocate for the collective gathering of ICG images from lymphedema patients to enhance dataset diversity and boost the accuracy of models dedicated to lymphatic vessel segmentation and foot keypoint detection. Second, while expanding the training datasets to enhance LVADet's performance is crucial, we should also consider implementing uncertainty quantification to facilitate real-time physician involvement during LVA procedures. Lastly, we suggest exploring the integration of a technique known as reinforcement learning from human feedback, which would allow models to learn directly from physician input.

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